
THE EVOLUTION OF MACHINE LEARNING AND AI

Week 2 Quiz Summary

Responsible AI: everything the quiz can ask, in plain English

How to use this page. Week 2 is examined in the **Week 6 quiz** (50% of the module). The quiz is **multiple choice**. Everything it can ask about Week 2 is in the **twelve points** below, one per part of the lecture.

Read the boxes (about 20 minutes). Do the questions (about 20 minutes). Mark them. Re-read only the points you got wrong.

1 Part 1. The twelve points

1.1 Why any of this matters

A weather forecast only describes the world. A model that decides who gets a loan, a job interview or bail **changes** it. Once a prediction hands something out or holds it back, it stops being a **measurement** and becomes an **intervention**. Most failures are not clever model bugs: the model worked and the **system around it** did not.

1.2 The seven places harm gets in

Think of a pipeline with seven stops, and a name for what goes wrong at each:

World (the past as it happened: historical injustice) → **Data** (who got sampled: under-representation) → **Labels** (what counts as truth: label bias) → **Objective** (what you optimise: proxy variables) → **Model** (what it can represent: aggregation) → **Threshold** (where you cut: error allocation) → **Deploy** (who acts on it: automation bias).

And there is a loop: **today's decisions become tomorrow's training data**.

1.3 Three different things called “bias”

Statistical bias: your estimate is consistently off target. **Inductive bias**: the assumptions a model needs in order to generalise at all, and it is a **good** thing. **Societal bias**: a group is systematically disadvantaged. A model with **no inductive bias is useless**; a model with **unexamined societal bias is dangerous**. Mixing the two up lets someone defend the bad kind by praising the good kind, so always say which one you mean.

1.4 The data, and who is missing

There is no neutral dataset, because there was no neutral world to photograph. So the question is never “is this data biased?” but “**biased how, by what process, and what does that do to this decision?**” **Gender Shades** (2018) tested a commercial face classifier: **0.8%** error for lighter-skinned men, **34.7%** for darker-skinned women. One average number hid a forty-fold gap. Hence the cheapest good habit in this module: **report accuracy per group, or do not report accuracy**. And note the tension: collecting more data on under-represented people improves fairness but cuts against consent. Write the tension down; do not pretend you solved it.

1.5 The label

“Ground truth” is a human judgement wearing a lab coat: somebody decided what counts as true. Two traps. **Selective labels**: you only see what happened to the people the old system let through, so your accuracy partly measures **agreement with the old policy**. **Proxies**: removing a protected attribute does not remove the information, because other columns stand in for it and a good learner finds them (Amazon's recruiting tool). **Blinding the model does not blind the data**, and it blinds **you**, because you can no longer measure the gap. You cannot fix what you refuse to measure.

1.6 The objective

You almost never optimise the thing you care about; you optimise something you can count. Care about health, count **spending**. Care about good employees, count **who was hired before**. **Goodhart's law**: once a measure becomes a target, it stops being a good measure. So write down the **gap** between what you care about and what you can count, and put that paragraph in the model card.

1.7 The model, and where you cut

A model reports the **average**, and averages hide small groups. Then you choose a **threshold**, the score above which you say yes. **Not choosing is choosing 0.5**: the default in the library is a policy that nobody decided and your name is on. Before you tune anything, answer in writing: **who bears a false positive, who bears a false negative, and are they the same people?** If they are different people, you are not doing statistics any more; you are handing out harm, and you need a reason.

1.8 Four fairness definitions, and why you cannot have them all

Demographic parity: each group is approved at the same rate. **Equal opportunity**: among people who really would repay, each group is approved at the same rate. **Equalised odds**: both the true-positive and the false-positive rates match. **Calibration**: a score of 0.7 means the same thing for everyone. The **impossibility result** (Kleinberg, Mullainathan and Raghavan; and separately Chouldechova, 2016 to 2017) proves you cannot have all of these at once unless the groups have equal base rates or your predictions are perfect.

COMPAS is the case. Among people who did **not** reoffend, **44.9%** of Black defendants and **23.5%** of white defendants had been labelled higher risk. Among those who **did** reoffend, **47.7%** of white and **28%** of Black defendants had been labelled lower risk. The vendor's reply was that the tool was **calibrated**, which was verifiably true. Both sides were right; they were measuring different things. So the theorem does not end the argument, it **moves it**: the answerable question is **"which definition does this decision need, on what account of the harm, and who signed off?"**

1.9 Deployment, and the loop

"A human reviews it" is only a real safeguard if the human **can disagree**: they need time, information, the model's uncertainty, and support from their manager when they overrule it. In the case files the harm was never exotic. It was an ordinary model multiplied by three things: **scale** (everyone at once), **speed** (faster than any human check) and **no way to appeal**. Those three are design choices, and they are yours.

1.10 Explanation

Two kinds. **Intrinsic**: the model is small enough to be its own explanation (a short list of rules, a scorecard). **Post-hoc**: a second model that describes the first, such as **LIME** (jiggle one case and fit a simple local model) or **SHAP** (share out the credit between features). Post-hoc explanations are **receipts that can be forged**: they can be unfaithful to the real model, can disagree with each other on the same case, and can be deliberately gamed. Rudin's position (2019): for high-stakes decisions, **build the simple interpretable model first and measure what the black box actually buys you**, then argue for it if it is worth it.

1.11 Accountability, and the law

Accountability belongs to **organisations and records**, not to models. If nobody's name is on the decision about what the target variable would be, that is not a philosophical puzzle; it is a **missing document**. The **EU AI Act** works like product safety, in four tiers: **unacceptable** (banned: social scoring, manipulative systems, untargeted face scraping), **high risk** (regulated: hiring, credit, education, essential services, policing), **transparency** (you must be told it is a machine; synthetic media must be marked) and **minimal** (most of what you will build). **GDPR** gives people rights over automated decisions, so build for the explanation you will owe. Two documents carry the load: a **datasheet** describing the dataset, and a **model card** describing the model, its intended use, its performance per group and the fairness definition you chose.

1.12 Where to fix it, and what each fix costs

Three places. **Pre-processing**: repair the data (reweight, resample). Treats the cause; but you may be required to keep the data intact. **In-processing**: build fairness into the training objective. Most principled; needs control of training. **Post-processing**: adjust thresholds or outputs per group. Works when you cannot retrain; often the legally riskiest in the EU. Now look at the pipeline order: post-processing patches stage 6 for a decision made at stage 4. **The cheapest fix is always the earliest one.**

2 Part 2. The seven questions for anything you build

You may be asked to **apply** these, so read each one as a test you could run.

#	Question
1	Who is in this data, and who is missing?
2	Who wrote the label, and what were they really recording?
3	What is the target a stand-in for, and where does the stand-in break?
4	Who bears a false positive, and who bears a false negative?
5	Which fairness definition am I committing to, and what am I giving up?
6	Can the person affected get an explanation and contest the outcome?
7	What happens when this output becomes next year's training data?

3 Part 3. Twenty practice questions

Same format as the quiz: one right answer each. Answers on the last page.

1. A bank approves loans with a model, and next year retrains it on the approvals it made. Which Week 2 idea is this?

- A. The feedback loop: today's decisions become tomorrow's training data
- B. Automation bias at the deploy stage
- C. Aggregation at the model stage
- D. Statistical bias in the estimator

2. A team deletes the gender column and says the model is now fair. What is wrong with that claim?

- A. Nothing, provided the column is deleted before training
- B. It works, but it reduces accuracy too much
- C. Other columns act as stand-ins, so the information remains, and now the disparity cannot be measured
- D. It is only valid if the model is a decision tree

3. A hospital predicts "patients who will need extra care" using past healthcare spending. What is the problem, and at which stage does it arise?

- A. The model cannot represent the relationship; the model stage
- B. The threshold is set too low; the threshold stage
- C. The data is too small; the data stage
- D. Spending is a stand-in for need, and people who were under-treated spent less; the objective stage

4. A model is reported at 94% accuracy. What is the single cheapest check that could change your view of it?

- A. A confusion matrix on the whole test set
- B. A second model for comparison
- C. Retraining with more epochs
- D. Accuracy broken down by group, because one average can hide a large gap

5. Journalists show a risk score has unequal false-positive rates between groups. The vendor shows the score is calibrated. Both verify their claims. What follows?

- A. The model is fair, since calibration is the stronger property
- B. One side must have made an arithmetic error
- C. Both can be true at once: the definitions conflict, so somebody must choose which one this decision needs
- D. The model is unfair, since unequal error rates always override calibration

6. You want: among applicants who really would repay, each group is approved at the same rate. Which definition is that?

- A. Demographic parity
- B. Calibration
- C. Equal opportunity
- D. Equalised odds

7. A team ships a classifier and leaves the threshold at the library default. What have they done?

- A. Maximised accuracy by definition
- B. Guaranteed equal error rates between groups
- C. Adopted a policy about who bears the errors, without deciding or recording it
- D. Nothing: 0.5 is a neutral choice

8. In a cancer screening tool, who bears a false negative, and why does the answer matter?

- A. The model's accuracy score; it matters for reporting
- B. The patient, whose disease is missed; the answer decides which errors you should be willing to trade
- C. Nobody, because a second test will catch it
- D. The hospital, which loses money; it matters for budgeting

9. A caseworker must review 60 model decisions an hour, sees no confidence score, and is questioned by her manager whenever she overrules the system. Is this human oversight?

- A. Yes: a human reviews every decision
- B. No: oversight is only a control when the reviewer can realistically disagree
- C. Yes, provided the reviews are logged
- D. No, but only because 60 an hour is too many

10. Two post-hoc explanation methods give different reasons for the same refusal. What does that tell you?

- A. One method has a bug and should be discarded
- B. The refusal was incorrect
- C. Explanations are descriptions of the model, not the model itself, and they can be unfaithful
- D. The model is unfair

11. A decision affects someone's liberty or livelihood. What is Rudin's advice, in practice?

- A. Use a black box, then attach SHAP explanations
- B. Always use the most accurate model available
- C. Fit the interpretable model first and measure what the black box actually gains; argue for the gap, do not assume it
- D. Avoid machine learning entirely for such decisions

12. A start-up sells a CV-screening tool to employers in the EU. Which risk tier applies, and what does that mean?

- A. High risk: risk management, data governance, logging, human oversight and conformity assessment
- B. Transparency: it must only disclose that it is a machine
- C. Unacceptable: it is banned outright
- D. Minimal: nothing is required

13. A target variable chosen badly in month one is causing disparities. The team responds by adjusting thresholds per group after training. What is the trade-off?

- A. It is the ideal fix, because it does not require retraining
- B. It is equivalent to fixing the target variable
- C. It patches stage 6 for a decision made at stage 4: workable, but the cheapest fix was the earliest one
- D. It is the cheapest fix, because it changes no data

14. A lender's model is trained only on people the old process approved. What is this, and what does high accuracy really show?

- A. Aggregation: accuracy hides small groups
- B. Automation bias: reviewers agreed with the model
- C. Label noise: accuracy is simply overstated
- D. Selective labels: accuracy partly measures agreement with the old policy

15. A dataset of voices contains few speakers with regional accents. Which stage is this, and what is the characteristic harm called?

- A. Labels: label bias
- B. Data: under-representation
- C. World: historical injustice
- D. Model: aggregation

16. “We cannot report disparities, because we do not collect the protected attribute.” What is the best engineering response?

- A. Infer the attribute from names and postcodes
- B. That is correct, and it protects the organisation
- C. Use accuracy alone, since it is group-neutral
- D. You cannot fix what you refuse to measure: plan lawful audit data at design time

17. What is a model card for, in one sentence?

- A. It documents the code so a successor can run it
- B. It reports the final accuracy figure for the buyer
- C. It proves the model complies with GDPR
- D. It records intended use, out-of-scope use, performance by group and the fairness choice made, so the decision has a name attached

18. A refused applicant asks why. The only answer the company can give is “the model scored you below the threshold”. On Week 2’s account, what is missing?

- A. An explanation naming which factors moved the score and in which direction, and a route to contest the decision
- B. A higher threshold
- C. A more accurate model
- D. A record of the training data

19. A moderation model wrongly removes a small number of posts. Which three factors turn that ordinary error rate into a serious harm?

- A. Cost, energy and data
- B. Scale, speed and no route of appeal
- C. Bias, variance and noise
- D. Accuracy, precision and recall

20. A colleague says: “All models are biased, so complaining about bias is naive.” Which distinction answers this?

- A. Statistical bias applies to estimates, not to models
- B. Only supervised models can be biased
- C. The complaint is naive, because bias cannot be measured
- D. Inductive bias is necessary for learning; societal bias is systematic disadvantage. They are different things

4 Answers

#	Answer	Why
1	A	Today's decisions become tomorrow's training data. Point 2.
2	C	Proxies keep the information, and blinding also blinds you to the gap. Point 5.
3	D	Spending stands in for need, and under-treated people spent less. Goodhart at the objective stage. Point 6.
4	D	Disaggregated evaluation: report accuracy per group, or do not report accuracy. Point 4.
5	C	The impossibility result: the definitions conflict, so the choice must be made and signed off. Point 8.
6	C	Equal true-positive rates between groups is equal opportunity. Point 8.
7	C	Not choosing is choosing 0.5, and it allocates the errors. Point 7.
8	B	The patient bears it; who bears which error decides the trade you are making. Point 7.
9	B	Oversight is only a control if the reviewer can realistically disagree. Point 9.
10	C	Post-hoc explanations describe the model and can be unfaithful or disagree. Point 10.
11	C	Fit the interpretable baseline and measure the gap before accepting a black box. Point 10.
12	A	Employment is a high-risk use under the EU AI Act. Point 11.
13	C	Post-processing patches stage 6 for a stage 4 decision; the cheapest fix is the earliest. Point 12.
14	D	Selective labels: you never see the refused cases. Point 5.
15	B	Who got sampled is the data stage; the harm is under-representation. Point 2.
16	D	You cannot fix what you refuse to measure. Points 4 and 5.
17	D	A model card is how a decision acquires a name. Point 11.
18	A	Build for the explanation you will owe: if you cannot say which factors moved the score, it is not deployable. Point 11.
19	B	Scale, speed and no appeal turn an ordinary error rate into harm. Point 9.
20	D	A model needs inductive bias; societal bias is a different thing with the same name. Point 3.

Scored under 15? Re-read the points for the questions you missed, then try again tomorrow. The answers say which point each question came from.